Introduction to Big Data

Agenda

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- History of Big Data
- Examples of Big Data
- Types of Big Data
- 3 V's in Big Data
- Jobs in Big Data

Introduction

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➢ What is Data?

The quantities, characters, or symbols on which operations are performed by a computer, which may be stored and transmitted in the form of electrical signals and recorded on magnetic, optical, or mechanical recording media.

➢ What is Big Data?

Big Data is also data but with a huge size. Big Data is a term used to describe a collection of data that is huge in volume and yet growing exponentially with time. In short such data is so large and complex that none of the traditional data management tools are able to store it or process it efficiently.

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Tabular Representation of Memory Sizes

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Name	Equal To	Size(In Bytes)
Bit	1 bit	1/8
Nibble	4 bits	1/2 (rare)
Byte	8 bits	1
Kilobyte	1024 bytes	1024
Megabyte	1, 024kilobytes	1, 048, 576
Gigabyte	1, 024 megabytes	1, 073, 741, 824
Terrabyte	1, 024 gigabytes	1, 099, 511, 627, 776
Petabyte	1, 024 terrabytes	1, 125, 899, 906, 842, 624
Exabyte	1, 024 petabytes	1, 152, 921, 504, 606, 846, 976
Zettabyte	1, 024 exabytes	1, 180, 591, 620, 717, 411, 303, 424
Yottabyte	1, 024 zettabytes	1, 208, 925, 819, 614, 629, 174, 706, 176

History of Big Data

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- **Early Data Collection (Pre-20th Century):** Before the 20th century, data collection was limited by the tools available. Data was primarily recorded on paper and collected manually.
- **Punch Cards (Late 19th to Early 20th Century):** The invention of punch cards by Herman Hollerith in the late 19th century revolutionized data processing. These cards were used to tabulate data for the U.S. Census of 1890, significantly speeding up the process.
- **Mainframe Computers (Mid-20th Century):** The advent of mainframe computers in the mid-20th century enabled organizations to store and process larger amounts of data electronically.

History of Big Data

- **Hadoop (Mid-2000s):** Hadoop, an open-source framework for distributed storage and processing of large datasets, was created by Doug Cutting and Mike Cafarella in the mid-2000s.
- **Web 2.0 and Social Media (Mid-2000s):** The popularity of Web 2.0 applications and social media platforms like Meta (Facebook) and X (Twitter) led to the rapid generation of user-generated content and social data.
- **Big Data Buzzword (Early 2010s):** The term "Big Data" gained mainstream attention in the early 2010s.
- **Current State (2024s):** Big Data continues to evolve, with new technologies and tools for data processing, storage, and analysis.

Examples of Big Data

- **Social Media:**

Social media platforms like Facebook, Twitter, and Instagram generate massive amounts of data every second.

- **Online Shopping:**

E-commerce websites like Amazon and Alibaba use Big Data to recommend products based on your browsing and purchase history.

- **Streaming Services:**

Platforms like Netflix and Spotify analyze your viewing and listening history to recommend movies, TV shows, or music based on your preferences

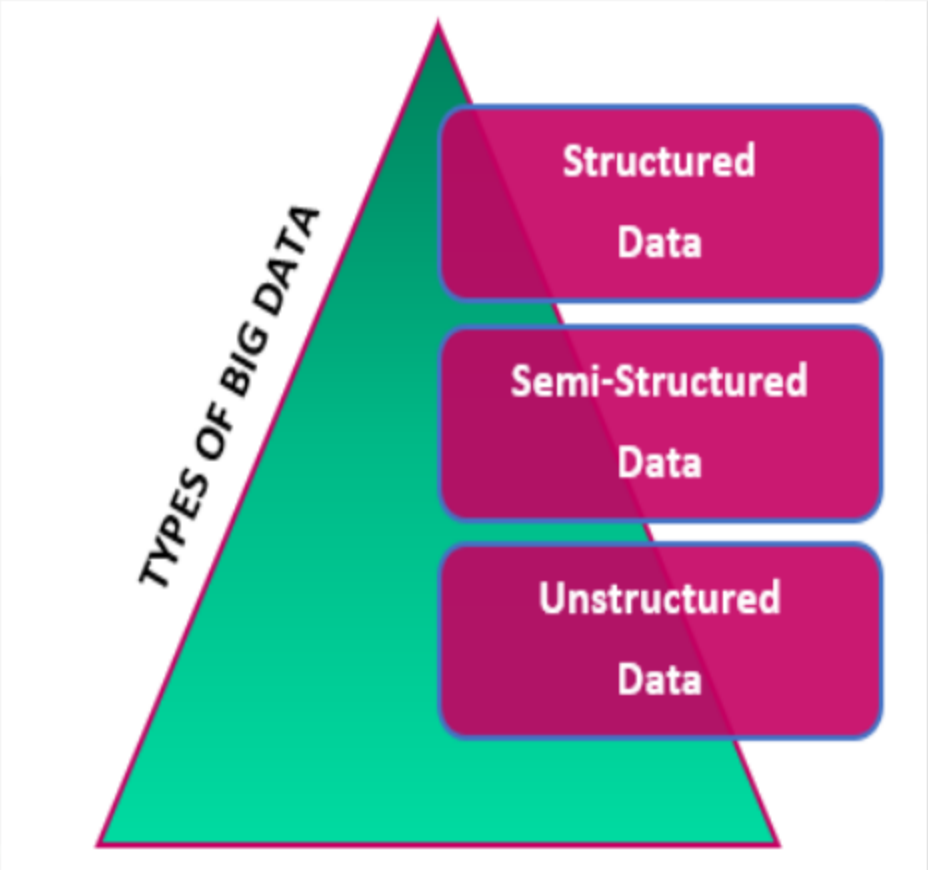
Types of Big Data

There are basically three types of Big Data:

1. Structured Data

2. Unstructured Data

3. Semi-Structured Data



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Types of Big Data

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- **Structured Data:**
- Structured data refers to data that is organized and formatted in a specific, well-defined manner, making it easily manageable, interpretable, and suitable for computer processing.
- This type of data conforms to a fixed schema, which defines the structure and relationships among the data elements.
- Examples:

Customer Database, Financial Records, Product Inventory, Students Record

Types of Big Data

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- **Unstructured Data:**
- Unstructured data refers to data that lacks a specific structure or format, making it more challenging to organize, analyze, and process using traditional data management tools and techniques.
- Unlike structured data, which is organized in rows and columns or follows a predefined schema, unstructured data does not adhere to a fixed structure and may come in various forms and formats.
- Examples:

Textual Documents, Images & Videos, Audio Recording, Social Media Content.

Types of Big Data

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- **Semi-Structured Data:**

- Semi-structured data is a type of data that falls between structured data and unstructured data.
- It possesses some structure or organization but does not adhere to a rigid schema or format like structured data.
- Examples:

Email, Sensor Data, NoSQL Databases

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3 V's in Big Data

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The concept of the "3 V's" in Big Data refers to three key characteristics or dimensions that help to define and understand the nature of big data.

The 3 V's are:

- **Volume:**

Big data is massive in scale. It includes data from various sources, such as social media, IoT (Internet of Things) devices, sensors, mobile devices, and more.

- **Velocity:**

The velocity refers to the speed at which data is generating. As technology is evolving the data is generating exponentially.

- **Variety:**

The variety means different types of big data. The data generated can be in any format such as images, audio, video, text, etc.

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Jobs In Big Data

Jobs in the field of big data can be found in a wide range of industries, including technology, finance, healthcare, retail, marketing, and more.

- Data Analyst
- Data Scientist
- Data Engineer
- Machine Learning Engineer
- Big Data Architect
- Database Administrator